

```
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score
from sklearn.metrics import confusion_matrix
from sklearn.metrics import classification_report
```

```
data = pd.read_csv("/content/Pass-Fail Data.csv")
```

```
print("First 5 records:")
print(data.head())
```

```
First 5 records:
  student_id  attendance_pct  homework_pct  midterm_score  \
0           1               95             92             88
1           2               88             85             79
2           3               60             55             58
3           4               72             70             65
4           5               40             45             50

  study_hours_per_week  pass
0                    12     1
1                    10     1
2                     4     0
3                     6     1
4                     3     0
```

```
print("\nDataset Information:")
print(data.info())
```

```
Dataset Information:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 100 entries, 0 to 99
Data columns (total 6 columns):
#   Column                Non-Null Count  Dtype
---  -
0   student_id            100 non-null   int64
1   attendance_pct         100 non-null   int64
2   homework_pct           100 non-null   int64
3   midterm_score          100 non-null   int64
4   study_hours_per_week   100 non-null   int64
5   pass                   100 non-null   int64
dtypes: int64(6)
memory usage: 4.8 KB
None
```

```
print("\nColumn Names:")
print(data.columns)
```

```
Column Names:
Index(['student_id', 'attendance_pct', 'homework_pct', 'midterm_score',
      'study_hours_per_week', 'pass'],
      dtype='object')
```

```
print("\nMissing Values:")
print(data.isnull().sum())
```

```
Missing Values:
student_id           0
attendance_pct       0
homework_pct         0
midterm_score        0
study_hours_per_week 0
pass                 0
dtype: int64
```

```
print("\nDuplicate Values:")
print(data.duplicated().sum())
```

```
Duplicate Values:
```

0

```
X = data[
    [
        "attendance_pct",
        "homework_pct",
        "midterm_score",
        "study_hours_per_week"
    ]
]

y = data["pass"]
```

```
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.20,
    random_state=42
)
```

```
print("\nTraining data size:", X_train.shape)
print("Testing data size:", X_test.shape)
```

```
Training data size: (80, 4)
Testing data size: (20, 4)
```

```
model = LogisticRegression()
```

```
model.fit(X_train, y_train)
```

```
▼ LogisticRegression ⓘ ?
LogisticRegression()
```

```
y_pred = model.predict(X_test)
```

```
print("\nActual Values:")
print(y_test.values)

print("\nPredicted Values:")
print(y_pred)
```

```
Actual Values:
[1 1 0 0 1 1 0 0 0 1 1 1 0 1 1 0 1 0 1 1]

Predicted Values:
[1 1 0 0 1 1 0 0 0 1 1 1 0 1 1 0 1 0 1 1]
```

```
accuracy = accuracy_score(y_test, y_pred)

print("\nAccuracy:", accuracy)
print("Accuracy Percentage:", accuracy * 100, "%")
```

```
Accuracy: 1.0
Accuracy Percentage: 100.0 %
```

```
cm = confusion_matrix(y_test, y_pred)

print("\nConfusion Matrix:")
print(cm)
```

```
Confusion Matrix:
[[ 8  0]
 [ 0 12]]
```

```
print("\nClassification Report:")
print(classification_report(y_test, y_pred))
```

```
Classification Report:
              precision    recall  f1-score   support

     0           1.00        1.00        1.00         8
     1           1.00        1.00        1.00        12

 accuracy          1.00
 macro avg          1.00
weighted avg          1.00
```

```
new_student = [[85, 80, 75, 10]]

prediction = model.predict(new_student)

print("\nNew Student Prediction:")

if prediction[0] == 1:
    print("Student will PASS")
else:
    print("Student will FAIL")
```

```
New Student Prediction:
Student will PASS
/usr/local/lib/python3.13/dist-packages/sklearn/utils/validation.py:2739: UserWarning: X does not have valid feature names, but
warnings.warn(
```

```
probability = model.predict_proba(new_student)

print("\nPrediction Probability:")
print("Fail Probability:", probability[0][0])
print("Pass Probability:", probability[0][1])
```

```
Prediction Probability:
Fail Probability: 5.149214388211476e-13
Pass Probability: 0.99999999999994851
/usr/local/lib/python3.13/dist-packages/sklearn/utils/validation.py:2739: UserWarning: X does not have valid feature names, but
warnings.warn(
```

```
print("\nModel Coefficients:")

for column, coefficient in zip(X.columns, model.coef_[0]):
    print(column, ":", coefficient)

print("\nIntercept:", model.intercept_[0])
```

```
Model Coefficients:
attendance_pct : 0.5916735823829831
homework_pct : 0.7354564535646119
midterm_score : 0.28241667584261326
study_hours_per_week : 0.31162436534831

Intercept: -105.13164332436409
```

